

GATE-DA — Artificial Intelligence

Mock Test – 1

Answer Key with Reasoning & NAT Solutions

Section A – Artificial Intelligence

Q1

Answer: A — [A, B], A on top

DFS pushes children onto the stack. To explore left-to-right (A before B), B must be pushed first and A pushed last, so A ends up on top: stack = [A, B] with A on top.

Q2

Answer: B

In UCS a node can later be reached via a cheaper path than the one first used to insert it into the frontier. Testing the goal only on POP (after the cheapest known cost is confirmed) guarantees optimality.

Q3

Answer: B — Not bipartite

2-colouring assigns alternating colours along edges. C and A are connected by an edge but both end up Red, which is a same-colour conflict on that edge $\Rightarrow$ the graph is not bipartite.

Q4

Answer: C — $O(V + E)$

Standard BFS visits every vertex once and examines every edge once (or twice for undirected graphs, which is still $O(E)$), giving $O(V + E)$.

Q5

Answer: A — $f(n) = 20$, the estimated total cost of the cheapest solution through n

$f(n) = g(n) + h(n) = 12 + 8 = 20$. This is A*'s estimate of the cheapest total path cost (start $\rightarrow$ n $\rightarrow$ goal) routed through node n.

Q6

Answer: C — 2

A MIN node propagates the minimum of its children's values: $\min(6, 2, 9) = 2$.

Q7

Answer: B — $O(4^6)$, identical to plain minimax

Alpha-beta pruning's effectiveness depends entirely on move ordering. With worst-case ordering, no branches get cut, so it degrades to the same complexity as plain minimax: $O(b^6) = O(4^6)$.

Q8

Answer: B — $\neg A$, by Modus Tollens

From $A \rightarrow E$ and observing $\neg E$, Modus Tollens lets us validly infer $\neg A$. (Modus Ponens would need A to conclude E, which is not what we're given.)

Q9

Answer: A — $A \perp C \mid B$

Skeleton edges: A–B, B–C, C–D, A–D, with D as a collider ($A \rightarrow D \leftarrow C$). Path A–B–C is a chain through B; conditioning on B blocks it. Path A–D–C is a collider at D; since D is NOT in the conditioning set, this path is also blocked. Both paths blocked $\Rightarrow A \perp C \mid B$ holds.

Q10

Answer: B — 5 variables {A,B,C,D,E}; 64 multiplications

The product factor $f_1 \cdot f_2 \cdot f_3$ (before summing out X) is defined over all variables appearing in any of the three factors: {X,A,B,C,D,E} — 6 binary variables $\Rightarrow 2^6 = 64$ entries, i.e. 64 multiplications to fill the table. After summing out X, the resulting factor is defined over the remaining 5 variables {A,B,C,D,E}.

Q11

Answer: C — prune after evaluating the 3rd child (value 2)

Start $\alpha=4$, $\beta=+\infty$. After 8: $\beta=\min(\infty,8)=8$, $\alpha(4) \geq \beta(8)$? No. After 5: $\beta=\min(8,5)=5$, $\alpha(4) \geq \beta(5)$? No. After 2: $\beta=\min(5,2)=2$, $\alpha(4) \geq \beta(2)$? Yes $\Rightarrow$ prune any remaining children.

Q12

Answer: B

Minimax guarantees the MAX player can secure AT LEAST the minimax value (7) against an optimal opponent. If the opponent plays sub-optimally, MAX may do even better — this does not violate the guarantee, which is a lower bound under optimal play, not an exact prediction.

Q13

Answer: B

Admissibility alone (without consistency) still guarantees A* tree-search optimality, but with graph search a node may first be reached via a costlier path and closed; if a cheaper path to that node is found later, A* must re-open and re-expand it to remain optimal.

Q14

Answer: A — straight-line (Euclidean) distance

Real road distance can never be shorter than the straight-line distance between two points (triangle inequality), so $h(n)$ = straight-line distance never overestimates the true cost $\Rightarrow$ admissible by construction. Scaling it up (option B), adding a buffer (option C), or using a proxy unrelated to true distance (option D) can all overestimate the true cost and break admissibility.

Q15

Answer: B — $O(2^{(1+\text{floor}(20/4))}) = O(2^6) = 64$

UCS's complexity bound is $O(b^{(1 + \text{floor}(C^*/\epsilon))})$. With $b=2$, $C^*=20$, $\epsilon=4$: $1 + \text{floor}(20/4) = 1+5 = 6$, giving $O(2^6) = 64$.

Q16

Answer: B — must store the entire frontier, exponential space

BFS keeps an entire level of the tree in memory at once. Since the frontier can grow as $O(b^d)$, memory (not just time) becomes the dominant practical bottleneck.

Q17

Answer: B — 121

BFS dequeues (expands) nodes level by level, testing each for goal-hood as it's removed from the frontier. With the goal as the very last node tested at depth d , every node from the root through the entire depth- d level must be dequeued/expanded first: $\text{total} = 1 + b + b^2 + b^3 + b^4 = 1 + 3 + 9 + 27 + 81 = 121$.

Q18

Answer: D — 0.450 (closest option to the computed value)

Using Bayes' rule: $P(C=1|A=1) = P(B=1) \cdot P(C=1|A=1, B=1) + P(B=0) \cdot P(C=1|A=1, B=0) = 0.6 \times 0.9 + 0.4 \times 0.5 = 0.74$.

$P(C=1|A=0) = 0.6 \times 0.4 + 0.4 \times 0.1 = 0.28$.

$P(C=1) = 0.3 \times 0.74 + 0.7 \times 0.28 = 0.222 + 0.196 = 0.418$.

$P(A=1|C=1) = P(A=1, C=1) / P(C=1) = 0.222 / 0.418 \approx 0.53$.

Q19

Answer: A

With a consistent heuristic, A^* expands nodes in non-decreasing order of f . Reasoning: when node N ($f=20$) is expanded, every node still sitting in the frontier already has $f \geq 20$ (since N was the minimum- f node chosen for expansion). Each child N produces also has $f(\text{child}) \geq f(N) = 20$, because consistency gives $h(N) \leq c(N, \text{child}) + h(\text{child})$, which rearranges to $f(\text{child}) \geq f(N)$. So immediately after N is expanded, the entire frontier is bounded below by 20, and this bound only gets reinforced at every subsequent step. Therefore no later pop can have $f=18$ if h is truly consistent.

Q20

Answer: B — $q \rightarrow p$

The converse of $p \rightarrow q$ swaps the antecedent and consequent: $q \rightarrow p$. ($\neg p \rightarrow \neg q$ is the inverse; $\neg q \rightarrow \neg p$ is the contrapositive, which is logically equivalent to the original, not its converse.)

Q21

Answer: 40

NAT solution: the goal is the last node expanded at depth $d=3$, meaning BFS dequeues/tests every node as it's removed from the frontier. So all nodes from the root through the full depth-3 level must be expanded: $3^0 + 3^1 + 3^2 + 3^3 = 1 + 3 + 9 + 27 = 40$.

Q22

Answer: 4

NAT solution: compare all $S \rightarrow G$ paths. $S \rightarrow A \rightarrow G = 3 + 2 = 5$. $S \rightarrow B \rightarrow G = 1 + 6 = 7$. $S \rightarrow B \rightarrow A \rightarrow G = 1 + 1 + 2 = 4$. The minimum is 4 (via $S \rightarrow B \rightarrow A \rightarrow G$).

Q23

Answer: 2

NAT solution: overestimate amount = $h(n) - h^*(n) = 9 - 7 = 2$. ($g(n)=18$ is irrelevant to this comparison.)

Q24

Answer: 3

NAT solution: Left MIN child = $\min(5, 8, 3) = 3$. Right MIN child = $\min(7, 2, 9) = 2$. Root is MAX, so value = $\max(3, 2) = 3$.

Q25

Answer: A, C

A is TRUE: with $h(n)=0$ everywhere, $f(n)=g(n)$ for all n , which is exactly UCS's ordering. C is TRUE: a consistent heuristic guarantees f -values never decrease along any path, so once a node is expanded with the cheapest possible g -value, it never needs reopening. B is FALSE: UCS corresponds to $h(n)=0$, not $h(n)=h^*(n)$ (that describes a 'perfect' heuristic, a different special case). D is FALSE: A^* with an admissible (but not necessarily perfect) heuristic typically expands fewer or equal nodes than UCS, never guaranteed strictly fewer in every case.

Q26

Answer: B, D

B is TRUE: the standard pruning cutoff condition is $\alpha \geq \beta$. D is TRUE: by definition, pruned branches are skipped entirely — never visited or evaluated. A is FALSE: alpha-beta is guaranteed to return the exact same root value as plain minimax; it only saves work, never changes the result. C is FALSE: the $O(b^{(m/2)})$ best case requires good/perfect move ordering; it is not achieved 'regardless of ordering'.

Q27

Answer: 512

NAT solution: $b^{(m/2)} = 8^{(6/2)} = 8^3 = 512$.

Q28

Answer: 16

NAT solution: eliminating A, B, C, D in order over the chain $A \rightarrow B \rightarrow C \rightarrow D \rightarrow E$ (all binary):

- Eliminate A: multiply $P(A) \cdot P(B|A)$, a table over $\{A, B\} \rightarrow 2^2 = 4$ multiplications.
- Eliminate B: multiply $f_1(B) \cdot P(C|B)$, a table over $\{B, C\} \rightarrow 4$ multiplications.
- Eliminate C: multiply $f_2(C) \cdot P(D|C)$, a table over $\{C, D\} \rightarrow 4$ multiplications.
- Eliminate D: multiply $f_3(D) \cdot P(E|D)$, a table over $\{D, E\} \rightarrow 4$ multiplications.

Total = $4+4+4+4 = 16$.

Q29

Answer: D

Pairwise conditional independencies ($X \perp Y|Z$ and $X \perp W|Z$) do not, by themselves, imply the joint statement $X \perp \{Y, W\}|Z$ — reaching that joint conclusion needs an extra assumption, in the spirit of the Intersection property (valid for strictly-positive distributions). Option D is the only one that acknowledges this extra premise is required rather than asserting the joint independence unconditionally, making it the best-supported choice.

Q30

Answer: 4600

NAT solution: $P(E=1) = 0.08$, so $P(E \neq 1) = 0.92$. Expected rejected samples = $N \times P(E \neq 1) = 5000 \times 0.92 = 4600$.

Section B – Aptitude

Q31

Answer: C — 20 m/s

Let train length = L . $(L+240)/24 = (L+120)/18$. Cross-multiplying: $18(L+240) = 24(L+120) \Rightarrow 18L+4320 = 24L+2880 \Rightarrow 6L = 1440 \Rightarrow L = 240$ m. Speed = $(240+240)/24 = 480/24 = 20$ m/s.

Q32

Answer: B — Lavish

'Parsimonious' means extremely frugal/stingy with money or resources. Its direct opposite is 'Lavish' (extravagant, generous to excess).

Q33

Answer: A — GMPXFS

The code shifts every letter forward by 1 (C→D, O→P, D→E, I→J, N→O, G→H). Applying the same +1 shift to FLOWER: F→G, L→M, O→P, W→X, E→F, R→S, giving GMPXFS.

Q34

Answer: B — 10 days

Rates: $(A+B)=1/12$, $(B+C)=1/15$, $(A+C)=1/20$. Sum: $2(A+B+C) = 1/12+1/15+1/20 = 5/60+4/60+3/60 = 12/60 = 1/5$. So $A+B+C = 1/10$ of the work per day $\Rightarrow$ together they finish in 10 days.

Q35

Answer: C — No valid conclusion follows

'All researchers are readers' places researchers wholly inside readers, but 'Some readers are writers' only guarantees overlap between readers and writers somewhere — not necessarily within the researcher subset. This is a classic syllogism trap; no conclusion about researchers and writers is logically forced.

Q36

Answer: A — 9

Let present ages be $3x$ and $5x$. After 9 years: $(3x+9)/(5x+9) = 3/4 \Rightarrow 4(3x+9) = 3(5x+9) \Rightarrow 12x+36 = 15x+27 \Rightarrow 3x = 9 \Rightarrow x = 3$. P's present age = $3x = 9$.

Q37

Answer: B — 24

A $4 \times 4 \times 4$ cube has 12 edges. Cubes with exactly 2 painted faces sit along edges, excluding the 2 corner cubes on each edge: $(4-2) = 2$ such cubes per edge $\times$ 12 edges = 24.

Q38

Answer: C — 20

Sanjay's position from the left = $N - 9 + 1 = N - 8$ (where N = total children). After swapping, Reema takes Sanjay's original spot, becoming 12th from the left: $N - 8 = 12 \Rightarrow N = 20$.

Q39

Answer: C — Atmospheric pressure

A thermometer measures temperature; analogously, a barometer measures atmospheric pressure.

Q40

Answer: B — 12.5%

Amount = $1.5P$, so Simple Interest = $0.5P$. $SI = P \times R \times T/100 \Rightarrow 0.5P = P \times R \times 4/100 \Rightarrow R = 0.5 \times 100/4 = 12.5\%$.